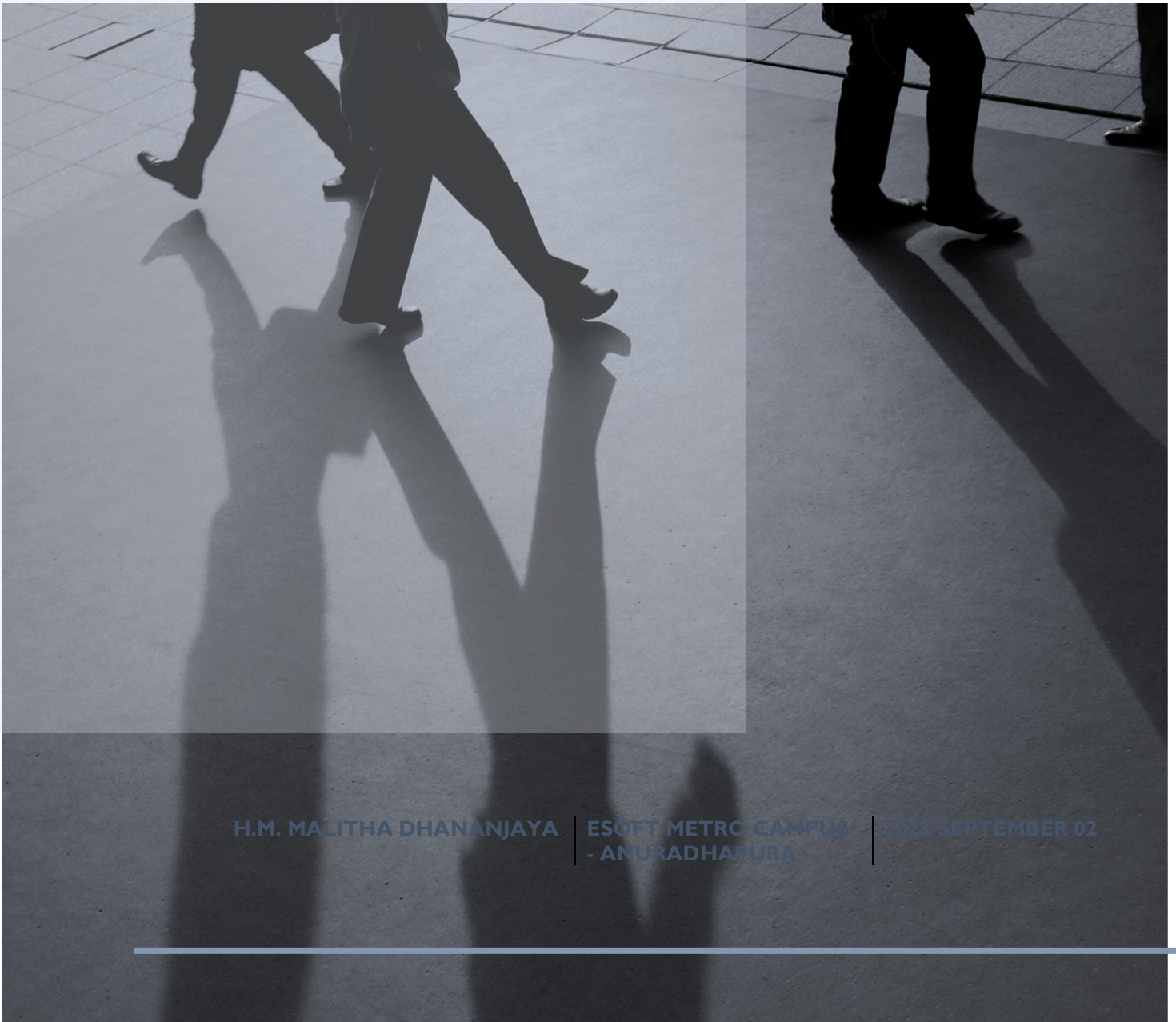


FINAL PROJECT DOCUMENTATION

STUDENT REGISTRATION SYSTEM



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- ANURADHAPURA

3025 SEPTEMBER 02

Final Project – Assessment Sheet	
Programme	L3 - Diploma in Information Technology
Assessor/Lecturer	
Unit	Final Project
Project Title	Student Registration System
Student's Name	H.M. Malitha Dhananjaya Herath
Student's Registration No	00271939
Submission Date	2025\09\01
Marks Achieved (%)	
Assessor/Lecturer Feedback	

.....
Lecturer signature



Student Signature

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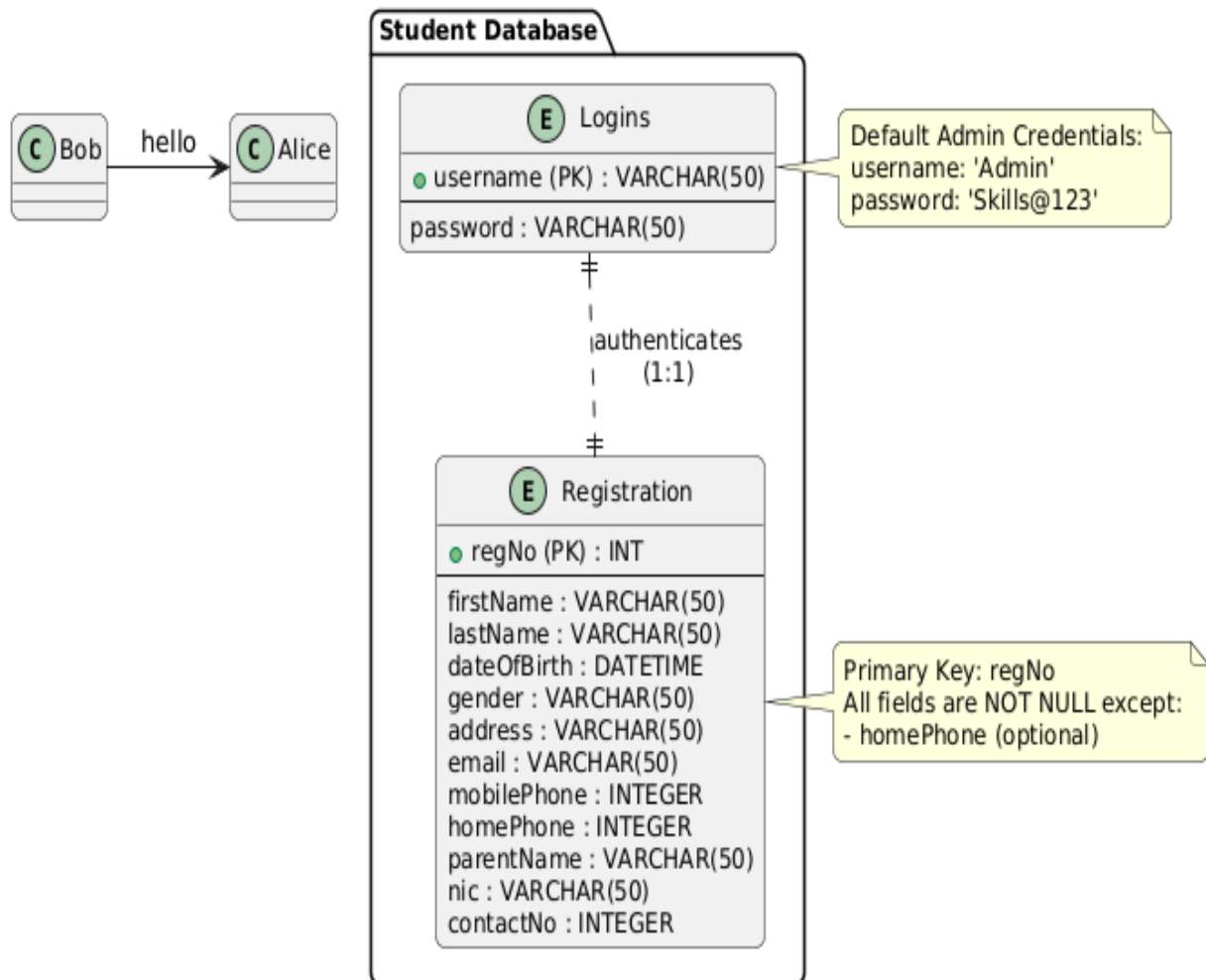
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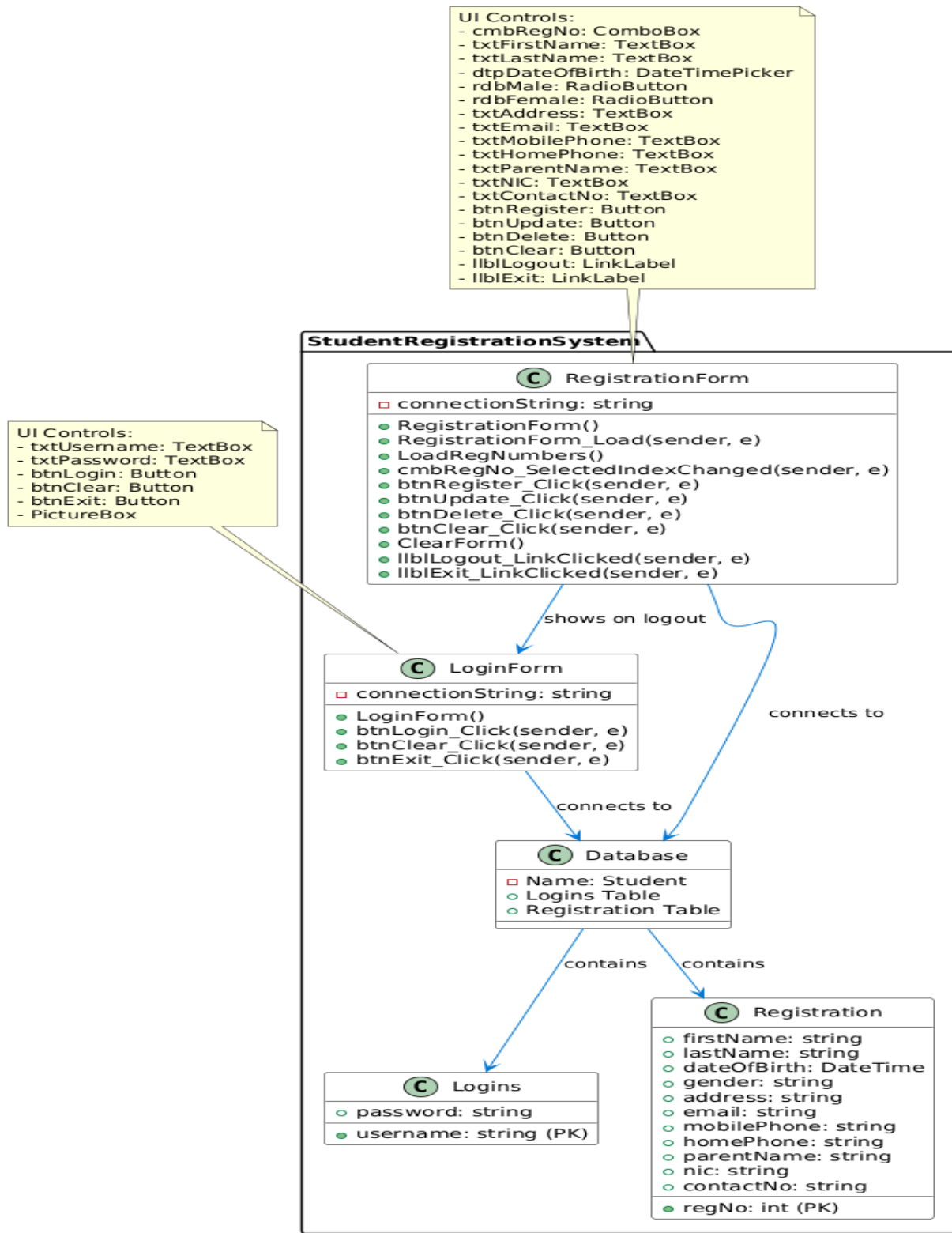
1. INTRODUCTION

This document provides a complete overview of the "Student Registration System" developed for Skills International School using C# .NET and Microsoft SQL Server. The system allows authorized administrators to manage student records through a secure login interface and a functional registration form that performs Create, Read, Update, and Delete (CRUD) operations.

ER Diagram for Student Registration System



Class Diagram for Student Registration System

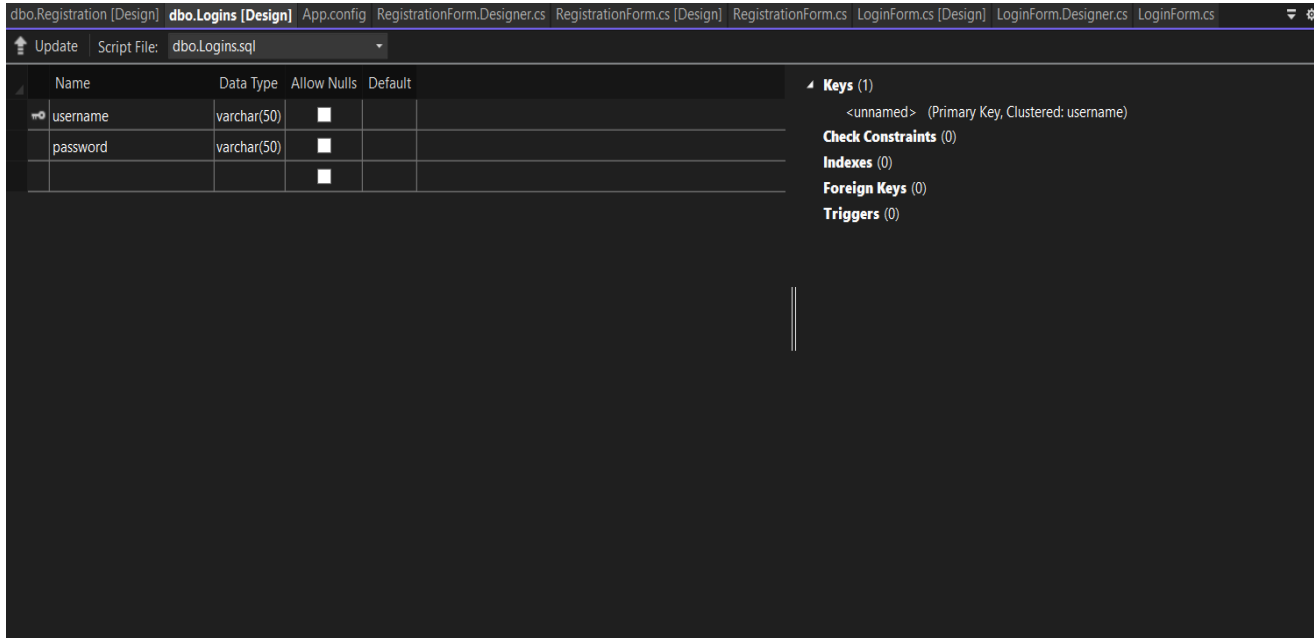


2. SYSTEM DESIGN

2.1 Database Design

The system uses a Microsoft SQL Server database named Student. It consists of two tables:

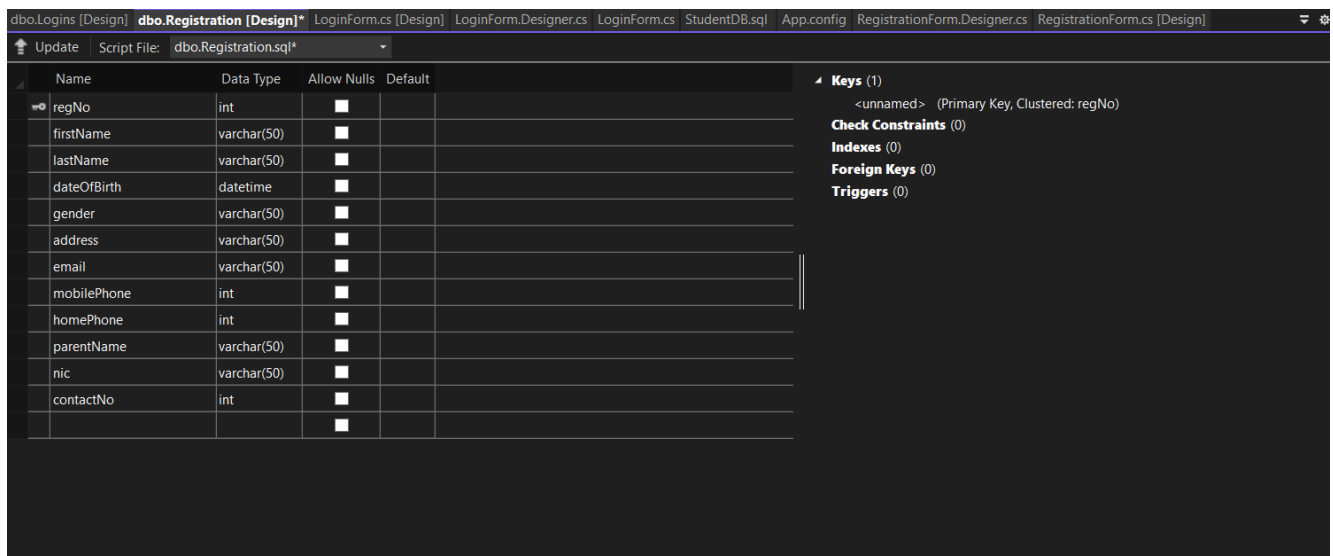
Table: Logins



The screenshot shows the SQL Server Enterprise Designer interface for the **dbo.Logins** table. The table has two columns: **username** (varchar(50)) and **password** (varchar(50)). The **username** column is marked as the primary key. The right-hand pane shows the table's properties: **Keys (1)** with a primary key named **<unnamed>** clustered on **username**; **Check Constraints (0)**; **Indexes (0)**; **Foreign Keys (0)**; and **Triggers (0)**.

Name	Data Type	Allow Nulls	Default
username	varchar(50)	☒	
password	varchar(50)	☒	

Table: Registration



The screenshot shows the SQL Server Enterprise Designer interface for the **dbo.Registration** table. The table has 14 columns: **regNo** (int), **firstName** (varchar(50)), **lastName** (varchar(50)), **dateOfBirth** (datetime), **gender** (varchar(50)), **address** (varchar(50)), **email** (varchar(50)), **mobilePhone** (int), **homePhone** (int), **parentName** (varchar(50)), **nic** (varchar(50)), and **contactNo** (int). The **regNo** column is marked as the primary key. The right-hand pane shows the table's properties: **Keys (1)** with a primary key named **<unnamed>** clustered on **regNo**; **Check Constraints (0)**; **Indexes (0)**; **Foreign Keys (0)**; and **Triggers (0)**.

Name	Data Type	Allow Nulls	Default
regNo	int	☒	
firstName	varchar(50)	☒	
lastName	varchar(50)	☒	
dateOfBirth	datetime	☒	
gender	varchar(50)	☒	
address	varchar(50)	☒	
email	varchar(50)	☒	
mobilePhone	int	☒	
homePhone	int	☒	
parentName	varchar(50)	☒	
nic	varchar(50)	☒	
contactNo	int	☒	

```
-- create Database
CREATE DATABASE StudentDB;

-- Create the Logins Table
USE StudentDB;
CREATE TABLE Logins (
    username VARCHAR(50) PRIMARY KEY,
    password VARCHAR(50) NOT NULL
);

-- Create the Registration Table
USE StudentDB;
CREATE TABLE Registration (
    regNo INT PRIMARY KEY,
    firstName VARCHAR(50) NOT NULL,
    lastName VARCHAR(50) NOT NULL,
    dateOfBirth DATETIME NOT NULL,
    gender VARCHAR(50) NOT NULL,
    address VARCHAR(50) NOT NULL,
    email VARCHAR(50) NOT NULL,
    mobilePhone INTEGER NOT NULL,
    homePhone INTEGER,
    parentName VARCHAR(50) NOT NULL,
    nic VARCHAR(50) NOT NULL,
    contactNo INTEGER NOT NULL
);

-- Insert the Default Admin User
USE StudentDB;
INSERT INTO Logins (username, password) VALUES ('Admin', 'Skills@123');
```

% No issues found

[Screenshot: SQL Server Management Studio (SSMS) script]

Connect to Database

```
Microsoft.Common.targets LoginForm.cs [Design] LoginForm.Designer.cs LoginForm.cs StudentDB.sql App.config RegistrationForm.Designer.cs RegistrationForm.cs [Design] RegistrationForm.cs
<?xml version="1.0" encoding="utf-8" ?>
<configuration>
  <connectionStrings>
    <add name="StudentDB"
      connectionString="Data Source=FREECY\SQLEXPRESS;Database=StudentDB;Trusted_Connection=True;TrustServerCertificate=True;"
      providerName="Microsoft.Data.SqlClient"/>
  </connectionStrings>
</configuration>
```

Registration Table

1

Login Table

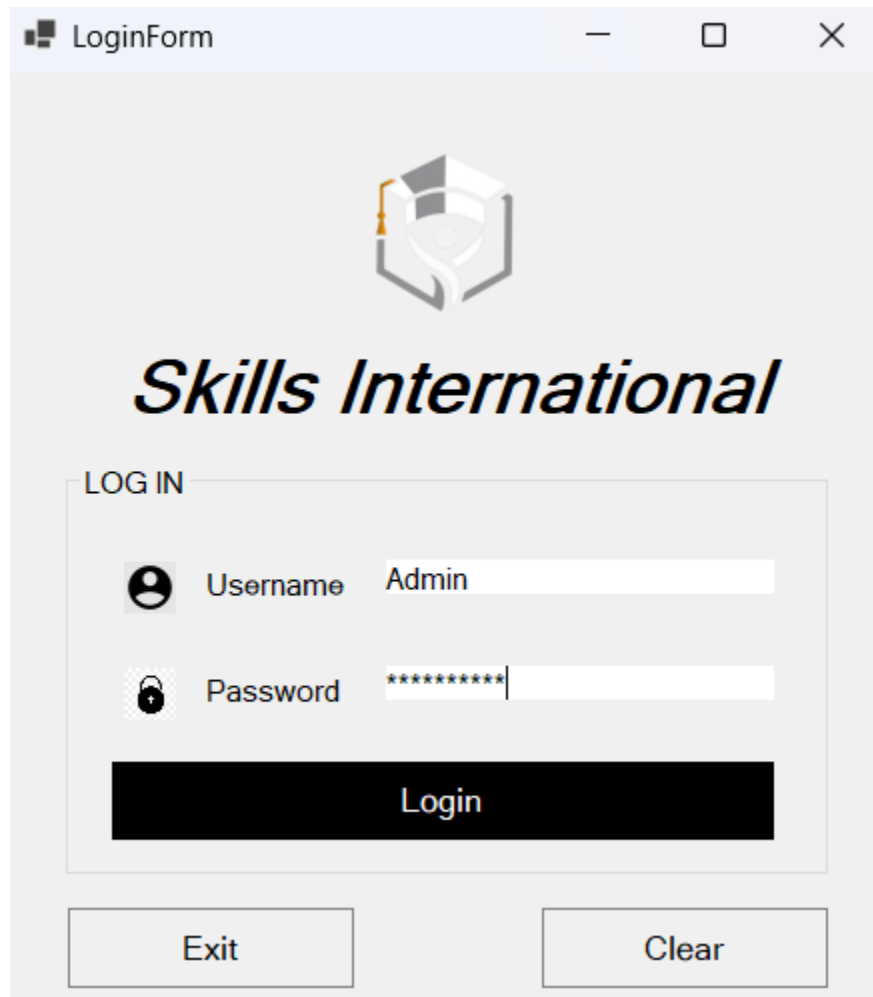
1 SELECT * FROM Logins;

%	✓ No issues found	▶
Results	Messages	
username	password	
Admin	Skills@123	

2.2 Application Design

The application consists of two main Windows Forms:

1. **Login Form:** Authenticates users.
2. **Registration Form:** Manages student data.



The screenshot displays a Windows Form titled "LoginForm". The interface features a logo at the top center, followed by the text "Skills International" in a large, bold, italicized font. Below this, there is a "LOG IN" section containing two input fields: "Username" with the value "Admin" and "Password" with masked characters "*****". A black "Login" button is positioned below the password field. At the bottom of the form, there are two buttons: "Exit" and "Clear".

Login Form Design

[Logout](#)

Skills International

Student Registration

Reg No

Basic Details

First Name

Last Name

Date of Birth

Gender ☒ Male ☐ Female

Contact Details

Address

Email

Mobile Phone Home Phone

Parent Details

Parent Name

NIC

Contact No

Register

Update

Clear

Delete

[Exit](#)

Registration Form

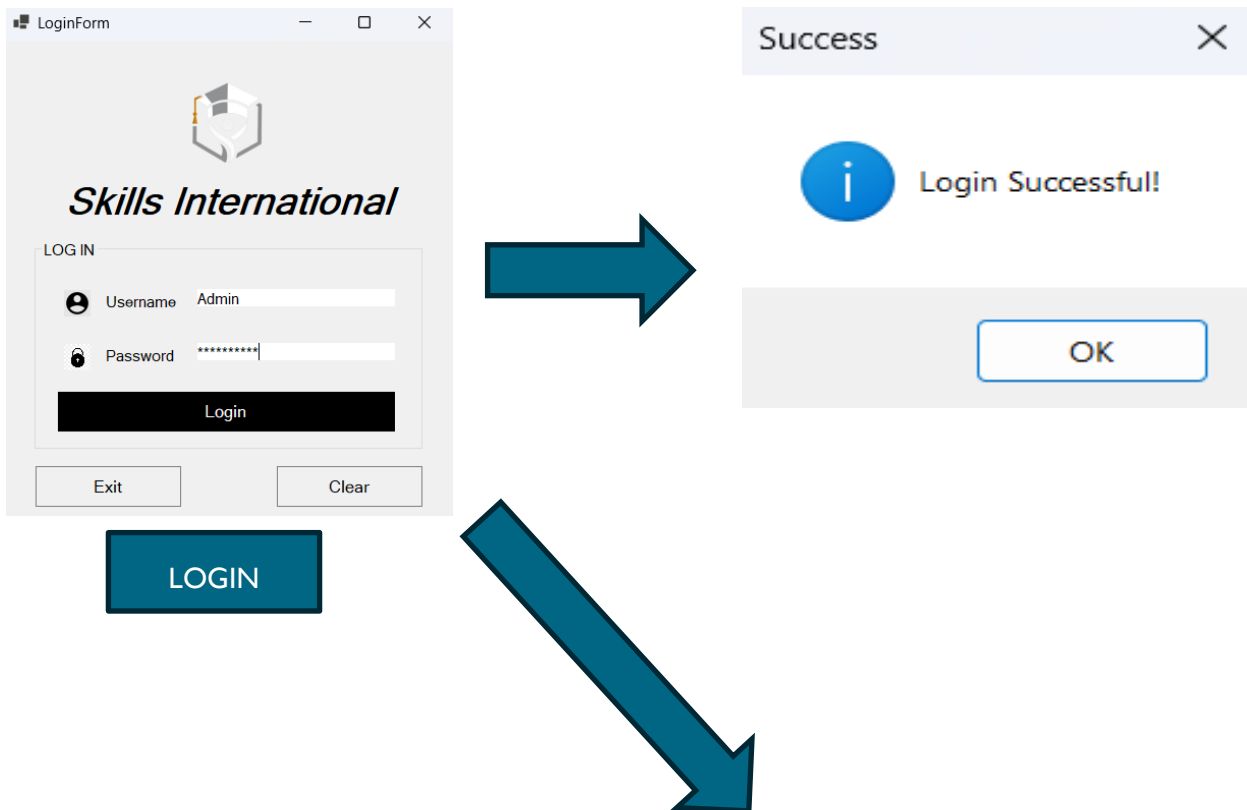
3. Implementation

3.1 Login Form (LoginForm.cs)

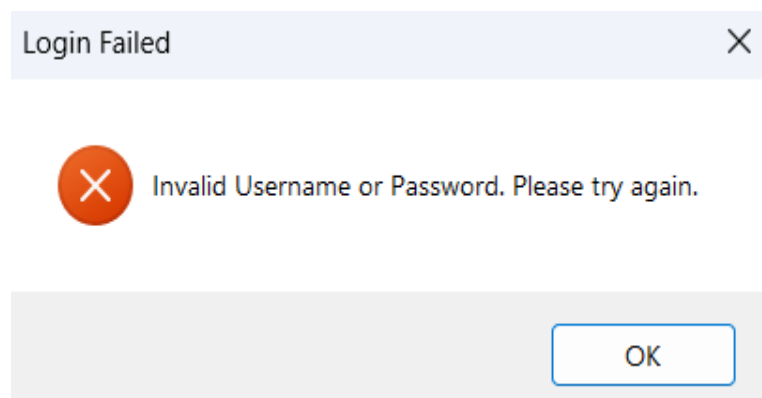
Design: The form was designed according to Figure I using Labels, TextBoxes, Buttons, and a PictureBox.

Key Functionality:

- Accepts username and password.
- Validates credentials against the Logins table in the database.
- **Login** button
- Provides **Clear** and **Exit** buttons.



Username or Password Error message:



Code Snippet: btnLogin_Click method

```
using System;
using Microsoft.Data.SqlClient;
using System.Configuration;
using System.Windows.Forms;

namespace StudentRegistrationSystem
{
    5 references
    public partial class LoginForm : Form
    {
        string connectionString = ConfigurationManager.ConnectionStrings["StudentDB"].ConnectionString;

        2 references
        public LoginForm()
        {
            InitializeComponent();
        }

        1 reference
        private void btnLogin_Click(object sender, EventArgs e)
        {
            using (SqlConnection conn = new SqlConnection(connectionString))
            {
                try
                {
                    conn.Open();
                    string query = "SELECT COUNT(1) FROM Logins WHERE username=@username AND password=@password";

                    using (SqlCommand cmd = new SqlCommand(query, conn))
                    {
                        cmd.Parameters.AddWithValue("@username", txtUsername.Text.Trim());
                        cmd.Parameters.AddWithValue("@password", txtPassword.Text.Trim());

                        int count = Convert.ToInt32(cmd.ExecuteScalar());

                        if (count == 1)
                        {
                            MessageBox.Show("Login Successful!", "Success", MessageBoxButtons.OK, MessageBoxIcon.Information);
                            this.Hide();
                            RegistrationForm regForm = new RegistrationForm();
                            regForm.Show();
                        }
                        else
                        {
                            MessageBox.Show("Invalid Username or Password. Please try again.", "Login Failed", MessageBoxButtons.OK, MessageBoxIcon.Error);
                        }
                    }
                }
                catch (Exception ex)
                {
                    MessageBox.Show("An error occurred: " + ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
                }
            }
        }
    }
}
```

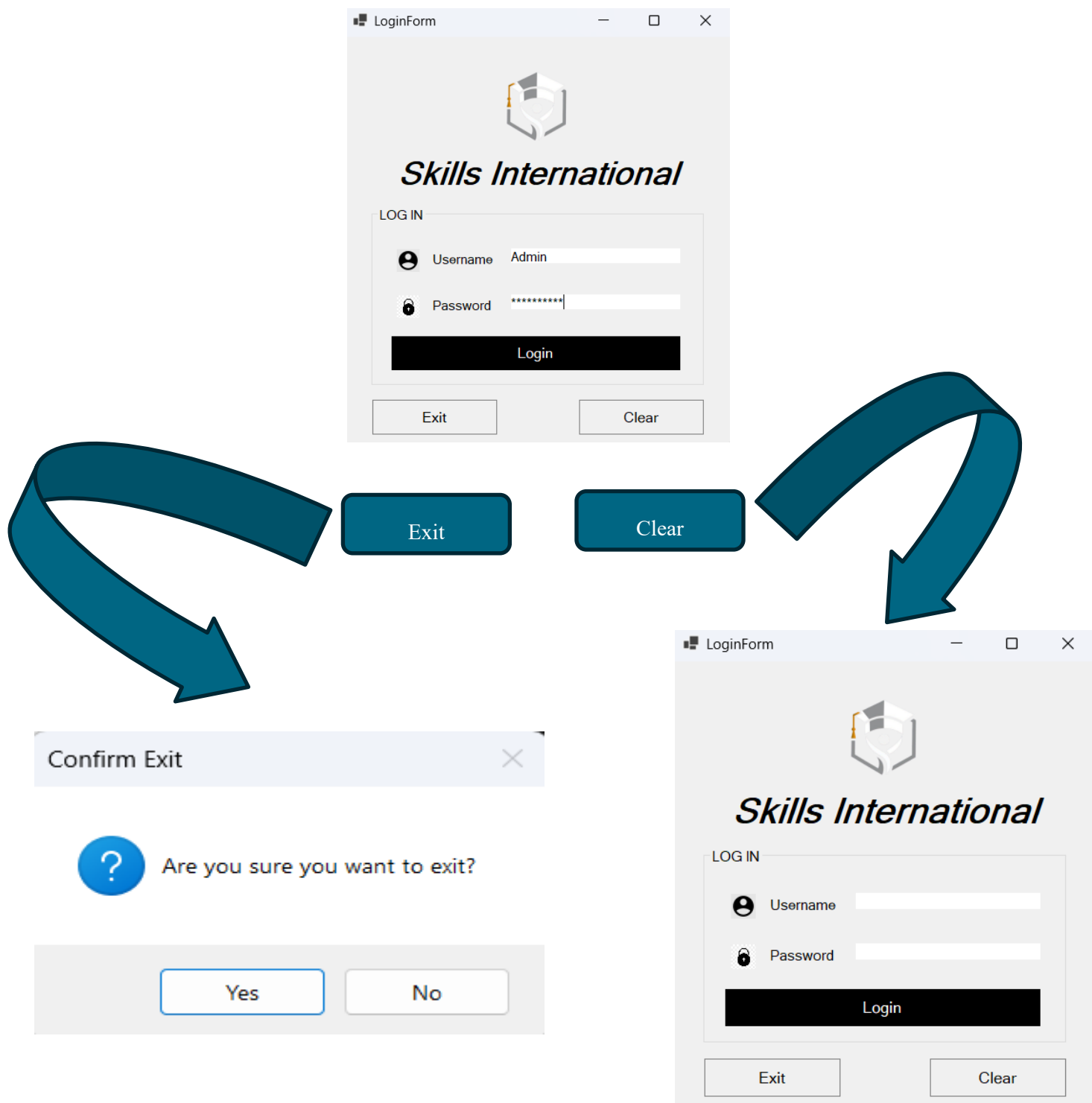
- Provides **Clear** and **Exit** buttons.

Code Snippet: btnClear and btnExit_Click method

```
1 reference
private void btnClear_Click(object sender, EventArgs e)
{
    txtUsername.Clear();
    txtPassword.Clear();
    txtUsername.Focus();
}

1 reference
private void btnExit_Click(object sender, EventArgs e)
{
    DialogResult result = MessageBox.Show("Are you sure you want to exit?", "Confirm Exit", MessageBoxButtons.YesNo, MessageBoxIcon.Question);
    if (result == DialogResult.Yes)
    {
        Application.Exit();
    }
}
```

Design View: **btnClear** and **btnExit_Click**



3.2 Registration Form (RegistrationForm.cs)

Design: The form was designed according to Figure II using various controls, including Combo Box, Text Boxes, Radio Buttons, DateTimePicker, and Buttons.

Key Functionality:

- **Register:** Inserts new student records into the database.
- **Update:** Modifies existing student records.
- **Delete:** Removes student records with confirmation.
- **Search:** Auto-fills the form when a regNo is selected from the dropdown.
- **Clear:** Resets all input fields.
- **Logout/Exit:** Navigates back to the login screen or exits the application.

The screenshot shows a Windows application window titled "RegistrationForm". Inside the window, the title "Skills International" is displayed in a large, bold, italicized font. To the left of the title is a blue link labeled "Logout". Below the title, the form is organized into sections. The "Student Registration" section at the top contains a "Reg No" dropdown menu with the value "00271939". Below this is the "Basic Details" section, which includes text boxes for "First Name" (filled with "malitha") and "Last Name" (filled with "dhananjaya"), a "Date of Birth" dropdown menu showing "Saturday , August 30, 2025", and "Gender" radio buttons with "Male" selected. The "Contact Details" section follows, with text boxes for "Address" (filled with "152,anuradhapura"), "Email" (filled with "acb@gmail"), "Mobile Phone" (filled with "11237854"), and "Home Phone" (filled with "1456121"). The "Parent Details" section at the bottom contains text boxes for "Parent Name" (filled with "abc"), "NIC" (filled with "123456789415"), and "Contact No" (filled with "12345668"). At the bottom of the form are four buttons: "Register", "Update", "Clear", and "Delete". In the bottom right corner of the window, there is a blue link labeled "Exit".

Code Snippet: Registerform

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using Microsoft.Data.SqlClient;
using System.Configuration;
using System.Windows.Forms;

namespace StudentRegistrationSystem
{
    4 references
    public partial class RegistrationForm : Form
    {
        string connectionString = ConfigurationManager.ConnectionStrings["StudentDB"].ConnectionString;

        1 reference
        public RegistrationForm()
        {
            InitializeComponent();
        }

        0 references
        private void RegistrationForm_Load(object sender, EventArgs e)
        {
            LoadRegNumbers();
            ClearForm();
        }

        4 references
        private void LoadRegNumbers()
        {
            using (SqlConnection conn = new SqlConnection(connectionString))
            {
                try
                {
                    conn.Open();
                    string query = "SELECT regNo FROM Registration";
                    SqlCommand cmd = new SqlCommand(query, conn);
                    SqlDataReader reader = cmd.ExecuteReader();

                    cmbRegNo.Items.Clear();
                    while (reader.Read())
                    {
                        cmbRegNo.Items.Add(reader["regNo"].ToString());
                    }
                    reader.Close();
                }
                catch (Exception ex)
                {
                    MessageBox.Show("Error loading registration numbers: " + ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
                }
            }
        }
    }
}
```

- **Register:** Inserts new student records into the database.

RegistrationForm

[Logout](#)

Skills International

Student Registration

Reg No

Basic Details

First Name

Last Name

Date of Birth

Gender ☒ Male ☐ Female

Contact Details

Address

Email

Mobile Phone Home Phone

Parent Details

Parent Name

NIC

Contact No

[Exit](#)

Error message:

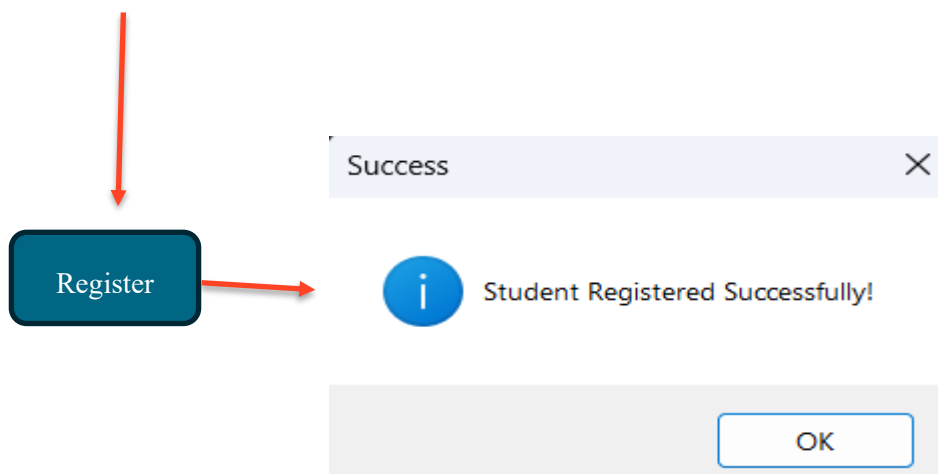
Error

 Please enter a Registration Number.

Registration Number already exists Error message:

Error

 This Registration Number already exists. Please use a different number.



Code Snippet: btnRegister

```
1 reference
private void btnRegister_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(cmbRegNo.Text))
    {
        MessageBox.Show("Please enter a Registration Number.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
        cmbRegNo.Focus();
        return;
    }

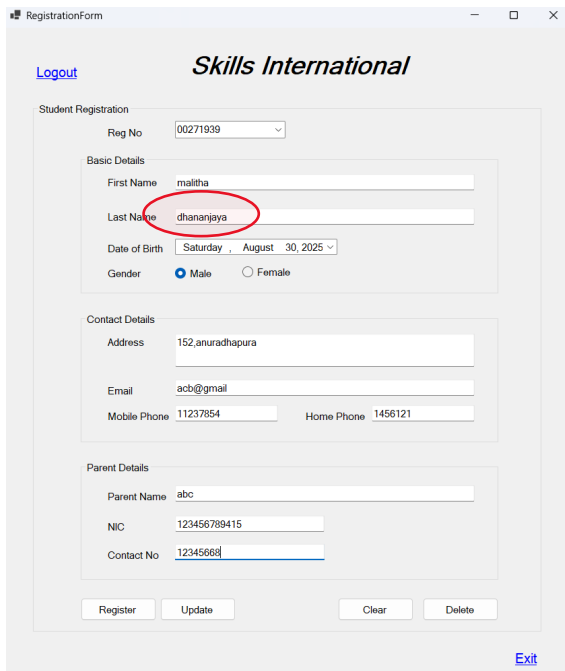
    int regNo = Convert.ToInt32(cmbRegNo.Text);
    string firstName = txtFirstName.Text;
    string lastName = txtLastName.Text;
    DateTime dateOfBirth = dtpDateOfBirth.Value;
    string gender = rdbMale.Checked ? "Male" : "Female";
    string address = textAddress.Text;
    string email = textEmail.Text;
    string mobilePhone = textMobilePhone.Text;
    string homePhone = textHomePhone.Text;
    string parentName = textParentName.Text;
    string nic = textNIC.Text;
    string contactNo = textContactNo.Text;

    using (SqlConnection conn = new SqlConnection(connectionString))
    {
        string query = @"INSERT INTO Registration
        (regNo, firstName, lastName, dateOfBirth, gender, address, email, mobilePhone, homePhone, parentName, nic, contactNo)
        VALUES
        (@regNo, @firstName, @lastName, @dob, @gender, @address, @email, @mobile, @home, @parent, @nic, @contact)";

        SqlCommand cmd = new SqlCommand(query, conn);
        cmd.Parameters.AddWithValue("@regNo", regNo);
        cmd.Parameters.AddWithValue("@firstName", firstName);
        cmd.Parameters.AddWithValue("@lastName", lastName);
        cmd.Parameters.AddWithValue("@dob", dateOfBirth);
        cmd.Parameters.AddWithValue("@gender", gender);
        cmd.Parameters.AddWithValue("@address", address);
        cmd.Parameters.AddWithValue("@email", email);
        cmd.Parameters.AddWithValue("@mobile", mobilePhone);
        cmd.Parameters.AddWithValue("@home", homePhone);
        cmd.Parameters.AddWithValue("@parent", parentName);
        cmd.Parameters.AddWithValue("@nic", nic);
        cmd.Parameters.AddWithValue("@contact", contactNo);

        try
        {
            conn.Open();
            cmd.ExecuteNonQuery();
            MessageBox.Show("Student Registered Successfully!", "Success", MessageBoxButtons.OK, MessageBoxIcon.Information);
            ClearForm();
            LoadRegNumbers();
        }
        catch (SqlException ex)
        {
            if (ex.Number == 2627)
            {
                MessageBox.Show("This Registration Number already exists. Please use a different number.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
            else
            {
                MessageBox.Show("Error saving data: " + ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
        }
        catch (Exception ex)
        {
            MessageBox.Show("An error occurred: " + ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
        }
    }
}
```

- **Update: Modifies existing student records.**



RegistrationForm

Logout Skills International

Student Registration

Reg No 00271939

Basic Details

First Name malitha

Last Name **dhananjaya**

Date of Birth Saturday, August 30, 2025

Gender ☒ Male ☐ Female

Contact Details

Address 152, anuradhapura

Email acb@gmail

Mobile Phone 11237854 Home Phone 1456121

Parent Details

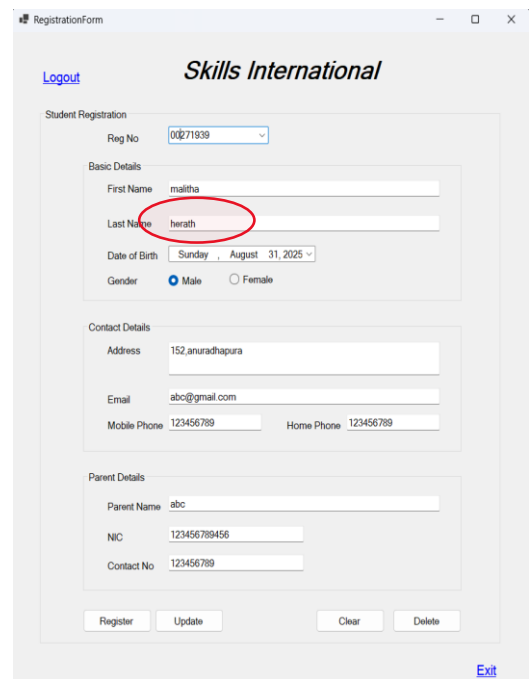
Parent Name abc

NIC 123456789415

Contact No 12345668

Register Update Clear Delete

Exit



RegistrationForm

Logout Skills International

Student Registration

Reg No 00271939

Basic Details

First Name malitha

Last Name **herath**

Date of Birth Sunday, August 31, 2025

Gender ☒ Male ☐ Female

Contact Details

Address 152, anuradhapura

Email acb@gmail.com

Mobile Phone 123456789 Home Phone 123456789

Parent Details

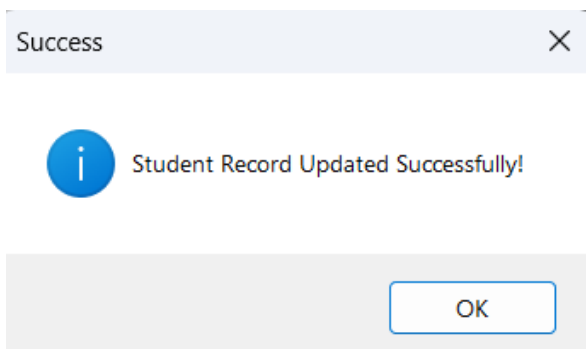
Parent Name abc

NIC 123456789456

Contact No 123456789

Register Update Clear Delete

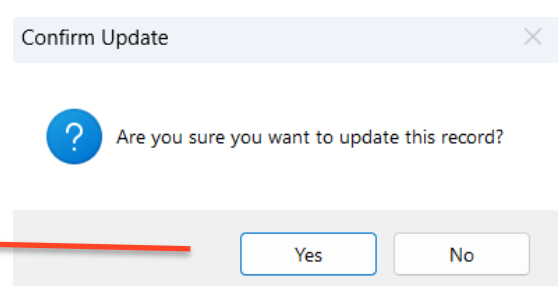
Exit



Success

Student Record Updated Successfully!

OK



Confirm Update

Are you sure you want to update this record?

Yes No

Error message:

Error



Please select a student to update.

OK

Code Snippet: btnUpdate

```
1 reference
private void btnUpdate_Click(object sender, EventArgs e)
{
    if (cmbRegNo.SelectedItem == null)
    {
        MessageBox.Show("Please select a student to update.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
        return;
    }

    DialogResult confirm = MessageBox.Show("Are you sure you want to update this record?", "Confirm Update", MessageBoxButtons.YesNo, MessageBoxIcon.Question);
    if (confirm == DialogResult.No) return;

    int regNo = Convert.ToInt32(cmbRegNo.SelectedItem.ToString());
    string firstName = txtFirstName.Text;
    string lastName = txtLastName.Text;
    DateTime dateOfBirth = dtpDateOfBirth.Value;
    string gender = rdbMale.Checked ? "Male" : "Female";
    string address = txtAddress.Text;
    string email = txtEmail.Text;
    string mobilePhone = txtMobilePhone.Text;
    string homePhone = txtHomePhone.Text;
    string parentName = txtParentName.Text;
    string nic = txtNIC.Text;
    string contactNo = txtContactNo.Text;

    using (SqlConnection conn = new SqlConnection(connectionString))
    {
        string query = @"UPDATE Registration SET
                        firstName=@firstName, lastName=@lastName, dateOfBirth=@dob, gender=@gender,
                        address=@address, email=@email, mobilePhone=@mobile, homePhone=@home,
                        parentName=@parent, nic=@nic, contactNo=@contact
                        WHERE regNo=@regNo";

        SqlCommand cmd = new SqlCommand(query, conn);
        cmd.Parameters.AddWithValue("@regNo", regNo);
        cmd.Parameters.AddWithValue("@firstName", firstName);
        cmd.Parameters.AddWithValue("@lastName", lastName);
        cmd.Parameters.AddWithValue("@dob", dateOfBirth);
        cmd.Parameters.AddWithValue("@gender", gender);
        cmd.Parameters.AddWithValue("@address", address);
        cmd.Parameters.AddWithValue("@email", email);
        cmd.Parameters.AddWithValue("@mobile", mobilePhone);
        cmd.Parameters.AddWithValue("@home", homePhone);
        cmd.Parameters.AddWithValue("@parent", parentName);
        cmd.Parameters.AddWithValue("@nic", nic);
        cmd.Parameters.AddWithValue("@contact", contactNo);

        try
        {
            conn.Open();
            int rowsAffected = cmd.ExecuteNonQuery();
            if (rowsAffected > 0)
            {
                MessageBox.Show("Student Record Updated Successfully!", "Success", MessageBoxButtons.OK, MessageBoxIcon.Information);
                ClearForm();
                LoadRegNumbers();
            }
        }
        catch (Exception ex)
        {
            MessageBox.Show("Error updating data: " + ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
        }
    }
}
```

- Delete: Removes student records with confirmation.

RegistrationForm

[Logout](#) **Skills International**

Student Registration

Reg No

Basic Details

First Name

Last Name

Date of Birth

Gender ☒ Male ☐ Female

Contact Details

Address

Email

Mobile Phone Home Phone

Parent Details

Parent Name

NIC

Contact No

[Exit](#)

Error message:

Error

 Please select a student to delete.

Delete

Confirm Delete

 Are you sure you want to DELETE this record? This action cannot be undone.

Success

 Student Record Deleted Successfully!

Code Snippet: btnDelete

```
private void btnDelete_Click(object sender, EventArgs e)
{
    if (cmbRegNo.SelectedItem == null)
    {
        MessageBox.Show("Please select a student to delete.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
        return;
    }

    DialogResult confirm = MessageBox.Show("Are you sure you want to DELETE this record? This action cannot be undone.", "Confirm Delete", MessageBoxButtons.YesNo, MessageBoxIcon.Warning);
    if (confirm == DialogResult.No) return;

    int regNo = Convert.ToInt32(cmbRegNo.SelectedItem.ToString());

    using (SqlConnection conn = new SqlConnection(connectionString))
    {
        string query = "DELETE FROM Registration WHERE regNo = @regNo";
        SqlCommand cmd = new SqlCommand(query, conn);
        cmd.Parameters.AddWithValue("@regNo", regNo);

        try
        {
            conn.Open();
            int rowsAffected = cmd.ExecuteNonQuery();
            if (rowsAffected > 0)
            {
                MessageBox.Show("Student Record Deleted Successfully!", "Success", MessageBoxButtons.OK, MessageBoxIcon.Information);
                ClearForm();
                LoadRegNumbers();
            }
        }
        catch (Exception ex)
        {
            MessageBox.Show("Error deleting data: " + ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
        }
    }
}
```

- **Search:** Auto-fills the form when a regNo is selected from the dropdown.

The screenshot shows the 'Skills International' RegistrationForm. The 'Student Registration' section has a 'Reg No' dropdown menu with '27456' and '271939' visible. Below it, the 'Basic Details' section includes 'First Name', 'Last Name', 'Date of Birth' (set to Sunday, August 31, 2025), and 'Gender' (Male/Female). The 'Contact Details' section has 'Address', 'Email', 'Mobile Phone', and 'Home Phone'. The 'Parent Details' section has 'Parent Name', 'NIC', and 'Contact No'. At the bottom are 'Register', 'Update', 'Clear', and 'Delete' buttons. A red circle highlights the 'Reg No' dropdown, and another red circle highlights the 'Date of Birth' calendar.

This screenshot is similar to the previous one, but the 'Date of Birth' calendar is expanded, showing a grid for August 2025. The calendar highlights the 31st. The 'Reg No' dropdown is now empty. A red circle highlights the expanded calendar.

Code Snippet: cmbRegN, DateTimePickerv, RadioButton(rdbFemale, rdbMale)

```

1 reference
private void cmbRegNo_SelectedIndexChanged(object sender, EventArgs e)
{
    if (cmbRegNo.SelectedItem != null)
    {
        string selectedRegNo = cmbRegNo.SelectedItem.ToString();
        using (SqlConnection conn = new SqlConnection(connectionString))
        {
            string query = "SELECT * FROM Registration WHERE regNo = @regNo";
            SqlCommand cmd = new SqlCommand(query, conn);
            cmd.Parameters.AddWithValue("@regNo", selectedRegNo);

            try
            {
                conn.Open();
                SqlDataReader reader = cmd.ExecuteReader();
                if (reader.Read())
                {
                    txtFirstName.Text = reader["firstName"].ToString();
                    txtLastName.Text = reader["lastName"].ToString();
                    dtpDateOfBirth.Value = Convert.ToDateTime(reader["dateOfBirth"]);

                    string gender = reader["gender"].ToString();
                    if (gender == "Male")
                    {
                        rdbMale.Checked = true;
                    }
                    else
                    {
                        rdbFemale.Checked = true;
                    }

                    txtAddress.Text = reader["address"].ToString();
                    txtEmail.Text = reader["email"].ToString();
                    txtMobilePhone.Text = reader["mobilePhone"].ToString();
                    txtHomePhone.Text = reader["homePhone"].ToString();
                    txtParentName.Text = reader["parentName"].ToString();
                    txtNIC.Text = reader["nic"].ToString();
                    txtContactNo.Text = reader["contactNo"].ToString();
                }
                reader.Close();
            }
            catch (Exception ex)
            {
                MessageBox.Show("Error loading student data: " + ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
        }
    }
}

```

- **Clear:** Resets all input fields.

RegistrationForm

[Logout](#) **Skills International**

Student Registration

Reg No: 00271939

Basic Details

First Name: malitha

Last Name: dhananjaya

Date of Birth: Saturday, August 30, 2025

Gender: ☒ Male ☐ Female

Contact Details

Address: 152, anuradhapura

Email: acb@gmail

Mobile Phone: 11237854 Home Phone: 1456121

Parent Details

Parent Name: abc

NIC: 123456789415

Contact No: 12345668

Register Update Clear Delete

[Exit](#)

RegistrationForm

[Logout](#) **Skills International**

Student Registration

Reg No:

Basic Details

First Name:

Last Name:

Date of Birth: Sunday, August 31, 2025

Gender: ☐ Male ☐ Female

Contact Details

Address:

Email:

Mobile Phone: Home Phone:

Parent Details

Parent Name:

NIC:

Contact No:

Register Update Clear Delete

[Exit](#)

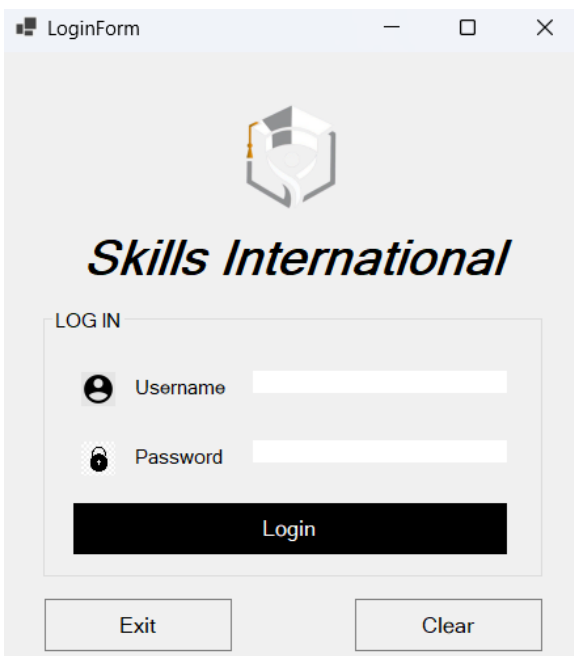
Clear

Code Snippet: btnClear

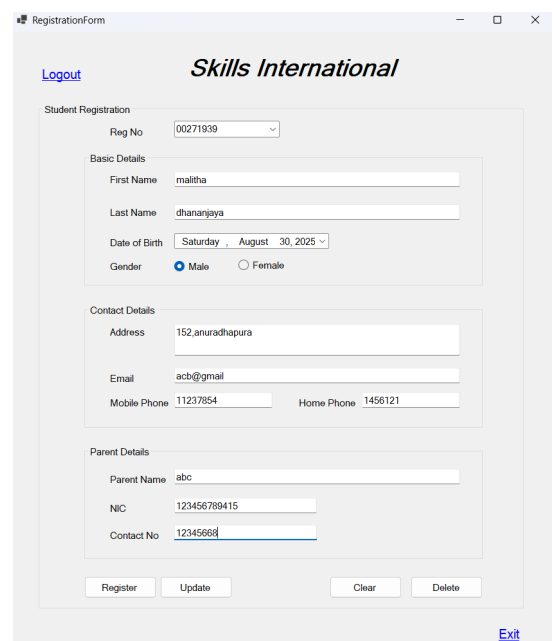
```
1 reference
private void btnClear_Click(object sender, EventArgs e)
{
    ClearForm();
}

5 references
private void ClearForm()
{
    cmbRegNo.SelectedItem = null;
    cmbRegNo.Text = "";
    txtFirstName.Clear();
    txtLastName.Clear();
    dtpDateOfBirth.Value = DateTime.Now;
    rdbMale.Checked = false;
    rdbFemale.Checked = false;
    textAddress.Clear();
    textEmail.Clear();
    textMobilePhone.Clear();
    textHomePhone.Clear();
    textParentName.Clear();
    textNIC.Clear();
    textContactNo.Clear();
    cmbRegNo.Focus();
}
```

- **Logout/Exit:** Navigates back to the login screen or exits the application.



[Logout](#)



RegistrationForm

[Logout](#) **Skills International**

Student Registration

Reg No

Basic Details

First Name

Last Name

Date of Birth

Gender ☒ Male ☐ Female

Contact Details

Address

Email

Mobile Phone Home Phone

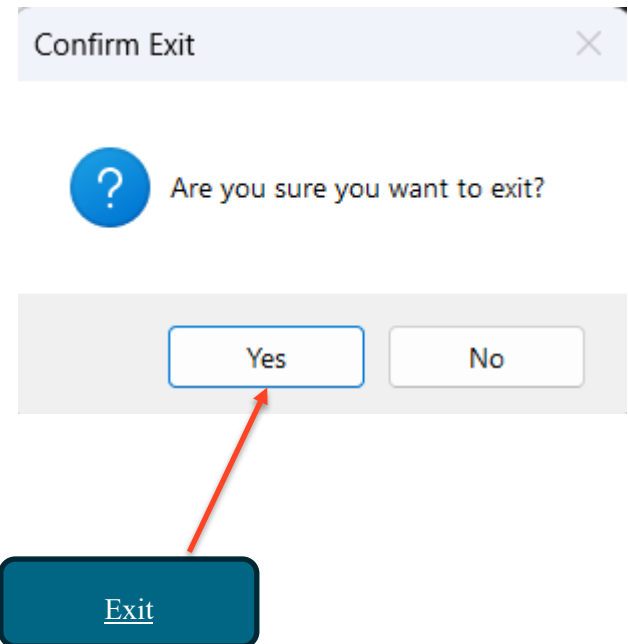
Parent Details

Parent Name

NIC

Contact No

[Exit](#)



Code Snippet: lblLogout, lblExit

```
1 reference
private void lblLogout_LinkClicked(object sender, LinkLabelLinkClickedEventArgs e)
{
    this.Hide();
    LoginForm loginForm = new LoginForm();
    loginForm.Show();
}

1 reference
private void lblExit_LinkClicked(object sender, LinkLabelLinkClickedEventArgs e)
{
    DialogResult result = MessageBox.Show("Are you sure you want to exit?", "Confirm Exit", MessageBoxButtons.YesNo, MessageBoxIcon.Question);
    if (result == DialogResult.Yes)
    {
        Application.Exit();
    }
}
```


4. Testing

A comprehensive test was conducted to ensure all functionalities work as intended.

Test Cases and Results:

Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Login with Valid Credentials	Username: Admin Password: Skills@123	Redirect to Registration Form	As Expected	Pass
2. Login with Invalid Credentials	Incorrect username/password	Error Message	As Expected	Pass
3. Register a New Student	Fill all required fields	Success Message, Data saved to DB	As Expected	Pass
4. Update a Student Record	Select Reg No, modify data, click Update	Success Message, Data updated in DB	As Expected	Pass
5. Delete a Student Record	Select Reg No, click Delete, confirm	Success Message, Data removed from DB	As Expected	Pass
6. Search Functionality	Select Reg No from the dropdown	Form auto-filled with student data	As Expected	Pass
7. Logout Functionality	Click the Logout link	Redirected to Login Form	As Expected	Pass
8. Exit Functionality	Click Exit link/button	Application closes after confirmation	As Expected	Pass

Build Success

```
Show output from: Build
Build started at 2:07 AM...
1>----- Build started: Project: StudentRegistrationSystem, Configuration: Debug Any CPU -----
1>StudentRegistrationSystem -> E:\esoft metro campus\Final Project\StudentRegistrationSystem\StudentRegistrationSystem\bin\Debug\net8.0-windows\StudentRegistrationSystem.dll
***** Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped *****
***** Build completed at 2:07 AM and took 01.391 seconds *****
```

5. Conclusion

The "Student Registration System" was successfully developed according to the project specifications. All required functionalities, including secure login and full CRUD operations for student data, have been implemented and tested. The application is robust, user-friendly, and effectively demonstrates the use of C# .NET with a SQL Server database.

6. References

1. *Project Brief - ESOF Metro Campus*
2. *Microsoft Documentation for C# and .NET*
3. *Microsoft SQL Server Documentation*

System Testing Video



Screen Recording
2025-09-01 021826.m

THANK YOU.....